

gRPC MACH Architecture

TCO Networking Costs Calculator

Point-to-Point Integration Pattern · Protocol Buffer IDL · AWS Egress Pricing

✔ File 'complexUnaryRPCPattern.proto' uploaded successfully!



Phase 4 — TCO Networking Costs Report

Full cost breakdown including architecture tactics, unit economics, and stakeholder summary.

SERVICE UNDER ANALYSIS

order-retriever-system

OMS

🔌 BUC1

⚡ 1000 base RPS

👥 231 End Users

📁 —

APPLIED ARCHITECTURE TACTICS

mTLS OAuth 2.0 + JWT



Applied Architecture Tactics & Patterns



INFORMATIONAL — NO RPS IMPACT

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
mTLS (mutual TLS)	0 reconnects/hr	Both client and server present X.509 certificates. CA validates both. 30: Max Payload overhead: bytes per message — RFC 8446 Because of RFC 8446 Section-5.2 * TLSCiphertext outer header: 5 bytes: * + 1-byte opaque_type=23 (uint8 ContentType) * + 2-byte legacy_record_version (uint16 ProtocolVersion) * + 2-byte length (uint16 length). *	None

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
		<i>TLSInnerPlaintext prefix (AAD): * + 1-byte Content type and no padding. * AEAD Pre-record nonce: RFC 8446 Section-5.3 * + 8-byte length (64 bits explicit portion added during encryption). * AEAD tag: RFC 5116 Section-5.1 * + 16-byte tag appended to encrypted_record for AEAD_AES_128_GCM ciphertext is exactly 16 octets longer than its corresponding plaintext . Handshake is connection-scoped — negligible RPS impact at this reconnect rate. AWS ACM certificates are free.</i>	
OAuth 2.0 + JWT — bearer token	Unary interceptor TTL 3600s 1 client(s)	<i>JWT Bearer header RFC 7515 Section-7.1 BASE64URL(UTF8(JWS Protected Header)) '.' * BASE64URL(JWS Payload) '.' * BASE64URL(JWS Signature) adds 650 bytes typical due to Payload overhead RFC 7519 §3 and RFC 9101: 650 bytes per message: Algorithm Standard claims RS256 Header 80 Payload (compact/std/ext) 200/400/600 Signature 344 Dots 2 Total 626/826/1026 → 650 typical . The token travels in every request's Authorization metadata header. Response messages carry no JWT overhead. Validation enforced by the configured gRPC interceptor.</i>	None



Security Byte Overhead Applied to Message Sizes

TLS OVERHEAD / MESSAGE	JWT OVERHEAD / REQUEST	EFFECTIVE REQUEST SIZE
30 bytes (typical)	650 bytes (typical)	8705 bytes
30 / bytes typical	300 / 650 / 1200 bytes (min/typical/max)	
EFFECTIVE RESPONSE SIZE		
824 bytes		

MESSAGE TYPE	PROTO SIZE (BYTES)	EFFECTIVE SIZE (BYTES)	REQUESTS / MONTH	THROUGHPUT (GB/MONTH)	TRANSFER COST (USD/MONTH)
Request	8025	8705	3B	21,013.7665 GB	—
Response	794	824	3B	1,989.1262 GB	\$ 179.02

 Total Cost of Ownership (TCO) Breakdown

Cost Component	DR / BC	Monthly Cost	% of TCO
AWS Egress (Response Transfer)	Networking	\$179.02	55.9%
Cloud Security Services	Disaster Recovery	\$68.00	21.2%
Containerized Environment Factors	Cloud Infra	\$73.00	22.8%
Total Cost of Ownership (TCO)		\$320.02/mo	100.0%

* % of TCO is calculated against the sum of ALL cost rows shown above (must total 100%). DR = Disaster Recovery | BC = Business Continuity.

 Unit Economics

TOTAL MONTHLY TCO (NETWORKING + CLOUD INFRA)

\$320.02

\$179.02 egress + \$141.00 cloud infra

 Per-Microservice Cost Metrics

TCO / MICROSERVICE /
MONTH

\$320.02

This service · full TCO

COST / REQUEST

\$0.00000012

2.592.000.000 req/mo

ANNUAL PROJECTION

\$3840.24

12× monthly TCO

 Per-End-User Cost Metrics (231 end users)

TCO / END USER / MONTH

\$1.3854

\$320.02 ÷ 231 users

TCO / END USER / DAY

\$0.046179

Monthly ÷ 30

BUSINESS USE CASE

—

Pattern cost attribution

Revenue, ARPU & ROI Analysis

Revenue model: The value entered in Phase 1 is the *expected monthly revenue per end user* for this service. Total monthly revenue = that value × number of end users. **ARPU** = revenue per user per month. **ROI** = (total revenue – TCO) ÷ TCO. **Break-even users** = minimum end users needed to cover full TCO at current ARPU. Net Margin per user = ARPU – TCO per user. Present to Finance and Leadership to frame architecture cost as a revenue-generating investment.

TOTAL MONTHLY REVENUE

\$3003

\$13.00/user × 231 users

ARPU / MONTH

\$13

Revenue per end user / month

NET MARGIN / USER / MO

\$11.6146

ARPU – TCO/user

MONTHLY ROI

838.4%

(Revenue – TCO) ÷ TCO

ANNUAL ROI

838.4%

12× monthly projection

BREAK-EVEN USERS

25

users to cover full TCO

REVENUE / \$1 INFRA

\$9.38

Revenue efficiency

ANNUAL REVENUE

\$36.036

12× monthly revenue

FinOps Architecture Notes & Disclaimers

1 – IP Address Planning (CIDR Sizing)

The more redundancy you build into AWS (Multi-AZ, replicas, NAT Gateways, ENIs), the more private IP addresses are consumed. Plan your VPC CIDR block generously (e.g. /16 = 65,536 IPs) and **leave enough address space within each subnet CIDR for auto-scaling headroom**. A /24 subnet provides only 251 usable IPs — exhausted quickly in a multi-tier, auto-scaled architecture.

2 – Multi-Tier Architecture in AWS Terms

Separating MACH components (frontend, API gateway, gRPC services, databases) into **different subnets** makes this a **multi-tier architecture** in AWS terms. Each tier needs dedicated subnets with its own Security Groups and NACLs.



3 — Intra-VPC Connectivity

Use **VPC Endpoints** (PrivateLink) to access AWS services without internet routing; **Transit Gateway** for cross-VPC connectivity; **VPC Flow Logs** for visibility. Each component in a separate subnet requires explicit route table entries.

* Outbound (response) data transfer cost uses AWS standard internet egress tiered pricing for US East (N. Virginia): first 10 TB at \$0.09/GB, next 40 TB at \$0.085/GB, next 100 TB at \$0.07/GB, beyond 150 TB at \$0.05/GB. Inbound (request) traffic is free. Effective sizes include TLS (RFC 8446) and JWT (RFC 7519) overhead where applicable. [AWS Pricing Calculator](#) ↗